

Supplementary Material for From Configurations to Claims: An Evidence Audit of Urban EV Charging Forecasting

Hongwei Chi

School of Computer Science and Technology, Hangzhou Dianzi University

Hangzhou 310018, Zhejiang, China

chihongwei@hdu.edu.cn

ORCID: [0009-0009-0393-999X](https://orcid.org/0009-0009-0393-999X)

Preprint version 0.9.1

Provisional authorship notice: Hongwei Chi is the confirmed first author. Additional authors and the corresponding author will be finalized before journal submission.

1 Full audit framework and claim grammar

Table 1: Portable fail-closed audit framework. Each transition requires an inspectable artifact; a failed precondition routes to *blocked* rather than to metric interpretation.

Stage	Required state	Minimum portable record	Fail-closed transition	Claim token released
1. Register	protocol fixed before aggregate evidence	data role, target, split, forecast origin, model/config ID, seed, endpoint, controls, gate, stop rule	missing or mutable field → BLOCKED	registered only
2. Execute	registered configuration	run ID, configuration fingerprint, code/environment identity, attempt state, immutable output paths	failure → terminal failed run; non-run → non-complete	executed or failed ; never reproduced by itself
3. Close	all expected cells terminal	expected/observed cardinality; target, order, time, spatial and mask hashes; prediction hashes; provenance links	incomplete or mismatched family → repair required	closed-unverified
4. Audit	closed matrix; metrics still locked	identity audit, boundary audit, control validity, repair parent/child hashes, prediction-change flag	any defect → invalidate affected formal family; non-selective rerun where selection could bias repair	audit-pass
5. Unlock	every registered audit passes	unlock authority, conditions, receipt hash, access counters, exact audited matrix hash	absent receipt or premature access → BLOCKED	metric-unlocked for the named role only
6. Decide	metric implementation and reference fixed	endpoint, reference, effect, uncertainty, consistency checks, conjunctive gate outcome	any required check fails → versioned hypothesis terminates	gate-pass or gate-fail ; subchecks stay typed
7. Write	decision and evidence chain retained	claim ID, authoritative artifact and hash, evaluation type, allowed wording, prohibited wording, scope limit	missing lineage → claim withheld	scoped manuscript statement

Table 2: Portable fail-closed audit framework (continued): typed claim grammar used to prevent diagnostic, proxy, engineering, or procedural evidence from being promoted to a stronger claim.

Evidence type	Required sentence grammar	Example permitted verb phrase	Promotion that remains prohibited
Oracle diagnostic	[target-dependent role] + [dataset boundary] + [reference] + [opportunity bound]	“indicates an infeasible upper bound”	“is deployable” or “can be recovered”
Proxy prediction	[proxy label] + [forecast-origin information set] + [discrimination/calibration]	“predicts relative expert advantage”	“improves forecast RMSE” or “identifies a cause”
Real-GT development/internal	[data role] + [reference] + [endpoint] + [effect/interval]	“improves internal rolling RMSE by $x\%$ ”	“generalizes”, “blind test”, or “state of the art”
Engineering measurement	[device + workload + precision + batch] + [endpoint] + [comparison]	“measured P50 was x ms on the frozen workload”	“is efficient” without a portable serving study
Conjunctive gate	[candidate] + [all required checks] + [pass/fail] + [authorized next action]	“failed the 1% point-effect requirement; stage stopped”	replacing the frozen question with a positive subcheck
Artifact-integrity event	[defect] + [pre-metric state] + [repair/audit] + [unlock outcome]	“blocked interpretation until identity audit passed”	“repair improved accuracy”
Procedural data boundary	[asset role] + [storage state] + [analytical-use state] + [threat-model limit]	“vault-materialized; analytical access after lock was zero”	“never materialized” or “cryptographically inaccessible”

2 Model identities and local adaptations

Table 3: Model identity, local adaptation, and claim scope. “Unknown/not claimed” is used when the frozen project artifacts do not establish a unique upstream identity or an authoritative field.

Paper label	Frozen implementation identity	Data/input contract and local changes	Role and explicit non-claim
UrbanEV CAPER	Existing exact-target expert predictions; legacy artifact label phase_only . The latency artifact reports 47,992 parameters. Upstream paper/repository checkpoint identity: unknown/not claimed .	UrbanEV panel, 168-hour project context and four horizons in the shared evaluation. The paper reuses predictions and does not reconstruct an unrecorded upstream training recipe or covariate contract.	Single expert, fixed-fusion component, router expert, and KD anchor. Not claimed as an exact replay of an external CAPER release or as a generally defined architecture.
Compact TimeXer-168	Local compact TimeXer-family artifact; 2,358,788 parameters in the latency serving artifact. Exact upstream repository revision for the UrbanEV artifact: unknown/not claimed .	Existing UrbanEV predictions with 168-hour context and shared target identity. No new expert training was performed for the audit chain.	Best evaluated single UrbanEV model within the frozen candidate set. Not “official upstream TimeXer”, universal best, or SOTA.
Global Fixed	Deterministic convex combination $\alpha_f \hat{y}^T + (1 - \alpha_f) \hat{y}^C$; one fold-level scalar pooled across horizons. Serving count 2,406,781 parameters includes both experts and scalar.	Fold 1 uses paired pre-test validation; folds 2–6 use earlier-fold OOF predictions only. Current-fold targets cannot fit α_f .	Selected fixed reference and distillation teacher. Not a cost-free ensemble, blind-test result, or cross-dataset claim.
Router v1	Local CAPER_TimeXer_ReliabilityGate; direct-loss MLP has hidden widths 32/16, horizon embedding 4, dropout 0.1, Huber loss, seed 42. Authoritative deployed parameter count: unknown/not claimed .	Expert outputs/disagreement plus 21 forecast-origin state features, rolling cross-fit; no feature timestamp exceeds the forecast origin.	Frozen learned-router comparison. It failed the 1% gate; no robustness or deployment qualification is claimed.
Chronos-2	Official amazon/chronos-2 snapshot resolved to commit 29ec3766d36d...e8498c ; chronos-forecasting 2.2.2 , Torch 2.5.1+cu121. Parameter count: unknown/not claimed .	Zero-shot FP32; 168-hour multivariate context; native Q0.5 output; no future covariates; predictions clipped to $[0, 1]$ for the frozen primary metric.	One teacher-qualification screen. Its failure is not evidence that foundation models generally fail for EV forecasting.
Residual student / aligned KD	Local CAPER-anchor plus shared DLinear residual branch; 338 trainable parameters. GT-only, aligned-KD, and shuffled-teacher variants share architecture, initialization, optimizer, data, epochs, and seed.	Occupancy-rate-only input, 168-hour lookback, 275 channels; aligned loss adds the frozen teacher-residual term with $\lambda = 0.5$.	Negative-gated distillation experiment. The aligned-vs-shuffled subcheck is not a passed improvement over GT-only and does not authorize latency or architecture search.
Paris CAPER	Paris_CAPER_phase_only , newly trained local adaptation; no UrbanEV checkpoint or fitted coefficient loaded. Authoritative parameter count: unknown/not claimed .	Development-only. Available, Charging, Passive, and Other map to available/active/unavailable; daily/weekly ordinal lags; target-history encoder plus station embedding replaces unavailable UrbanEV channels.	Paris single-model candidate. Must not be named unqualified CAPER or treated as an exact UrbanEV replay.
Paris TimeXer	TimeXer_local_audited_compact_L1 68 ; local source SHA-256 begins 563d4ce4304f ; model class source matches the workspace copy. Upstream commit is unknown/not claimed .	Retrained from initialization on Paris development; compact profile ($d = 128$, four heads, two layers); external train-only scaler, masked loss, direct endpoint, no time markers.	Paris single-model candidate. Explicitly a local audited adaptation, not an exact official upstream implementation.
Paris strict teacher	Fold-specific convex combination of the two Paris local adaptations; no separately trained joint model and no protected-target input.	Fold 1 α from pre-evaluation validation; later folds from prior-fold OOF predictions; shared station, mask, target, and timestamp identities.	Development-only teacher qualification. It failed the point-effect gate; student training and formal/protected evaluation were not authorized.

3 Research-question to claim map

Table 4: Research-question map from operational definition to bounded answer.

RQ	Operational question	Evidence and analysis	Direct result	Bounded answer
RQ1	How many frozen configurations are exact, disclosure-limited, trend-only, underidentified, failed, or non-complete?	1,224-row status registry; deterministic S1–S6 classifier; execution state, source-cell mapping, fold coverage, and worst disclosed-metric grade.	865 artifacts complete; S1=156, S2=216, S3=122, S4=371, S5=19, S6=340.	Completion is broader than exact reproduction in this inventory. Counts are not a field-wide reproducibility estimate or a model ranking.
RQ2	How much oracle complementarity survives feasible prequential fusion and learned routing?	Target-dependent oracle; expert residual diagnostics; forecast-origin proxy classifier; strict Global/Horizon Fixed controls; direct-loss and secondary router variants.	Oracle +16.926% versus TimeXer; Global Fixed +3.178%; primary router +0.443% versus Global Fixed.	A material fixed-fusion gain survives, but only a small additional router signal passes to the real-GT stage and it fails the frozen effect gate.
RQ3	What changes after synchronous latency and the minimum-effect rule are enforced?	RTX 4060 FP32 batch 1/8/32 benchmark; batch-1 primary endpoint; router, Chronos-2, KD, and Paris conjunctive gates.	Global Fixed has +48.86% batch-1 P50 versus TimeXer; later gains are +0.443%, −3.048%, +0.119%, and +0.847%.	The accuracy leader has a measured serving cost; positive or consistent subchecks do not automatically qualify the next engineering stage.
RQ4	How do fail-closed audits and stopping rules alter admissible claims?	Metric lock; non-selective control rerun; 32/32 Paris prediction closure; unchanged-prediction repair; versioned termination; procedural vault guard.	Three identity shuffles were excluded before metrics; 10,236 masked target entries were restored without prediction changes; analytical access after the Paris lock remained zero.	Integrity controls changed what could be interpreted and stopped failed versions. They do not prove higher accuracy, hardened security, or external validity.

A Reproducibility and Claim-Control Appendix

This appendix exports the audit interface behind the case study. It is deliberately documentary: it reuses frozen main-workspace artifacts, performs no model training or inference, recomputes no experiment metric, runs no new bootstrap or threshold sensitivity, and does not read Paris formal or protected target content. The tables distinguish facts present in the artifacts from portable recommendations. When a field cannot be established from a frozen artifact it is reported as *unknown/not claimed*; it is not reconstructed by assumption.

The appendix also does not supply evidence that the study did not collect. In particular, no independent-coder classification exercise, inter-rater agreement statistic, alternative bootstrap block-length analysis, or new Global Fixed uncertainty interval is claimed. The main text’s 0.5/1/2% threshold counterfactual is arithmetic context only; it does not recompute metrics or alter a decision. The S1–S6 registry is a deterministic single-pipeline classification of one inventory. Historical project records used a review label for what was actually a separately instantiated automated differential artifact audit, not a blinded external human replication or an inter-rater study.

A.1 Deterministic S1–S6 decision tree

The classifier in `experiments/00_registry_reclassification/scripts/rebuild_status_registry.py` applies the following ordered rules to each frozen manifest row. Order is part of the definition; a later numerical rule cannot overwrite an earlier execution fact.

- D1.** If `execution_status=failed`, assign S5 and require any future attempt to use a new versioned retry identifier.
- D2.** Otherwise, if `execution_status` is not `complete`, assign S6 and require any future execution to use a new versioned run identifier.
- D3.** Otherwise, if the row maps to UrbanEV source Table 4, assign S4 because the source paper and repository do not identify the disclosed node uniquely.
- D4.** Otherwise, if no source comparison cell can be mapped uniquely, assign S4.
- D5.** Otherwise, collect the disclosed RMSE, MAPE, RAE, and MAE grades and take the worst valid grade. The frozen bands are exact (relative error no greater than 2%), near (2–5%), and trend (greater than 5%); an empty valid-grade set defaults to trend when a comparison record exists.
- D6.** If comparison fold coverage is below six, assign S3. Otherwise, if the worst grade is trend, assign S3.
- D7.** Otherwise, if the track is `official` and the worst grade is exact, assign S1. Every other completed, uniquely mapped, non-trend row is S2.

Thus the executable priority is $S5/S6 \rightarrow S4 \rightarrow S1/S2/S3$. A completion mark indicates process completion only; it is not used as a synonym for S1. The following examples are the first row of each final status in the authoritative

CSV. The `classification_reason` text is an English rendering of the Chinese source field so that the appendix compiles under the paper’s Latin-font setup.

Table 5: One frozen registry example per evidence status. Full run identifiers and reasons are retained to make the classification path inspectable.

State	Example ID	Reason
S1	<code>official-compat-v1-t3-lo-occ-3-1-region-None-42-20-eae54b2d</code>	Official-compatible protocol is identifiable, all six folds are complete, and the relative error of each of four disclosed metrics is no greater than 2%.
S2	<code>audited-t3-lo-occ-3-1-region-None-42-20-efalea0d</code>	Six folds and the main protocol align, but an audited correction is present or at least one metric lies in the 2–5% approximation band.
S3	<code>audited-t3-ar-occ-3-1-region-None-42-20-3b0793b0</code>	At least one paper-disclosed metric has relative error above 5%; only trend-level evidence is retained.
S4	<code>audited-t4-lo-occ-3-1-194-None-42-20-42da10db</code>	Neither the paper nor repository discloses the Table 4 node identity; node 194 is a predeclared reverse-identification candidate, not a unique truth.
S5	<code>audited-t3-lstm-occ-6-3-region-None-42-20-c735895e</code>	The frozen configuration was executed but failed; the failure artifact is retained and cannot contribute a paper number.
S6	<code>audited-t3-TimesNet-occ-3-3-region-None-42-1-1687eb9d</code>	The configuration is frozen in the manifest but was not executed or registered after accumulated failures in the same model batch.

A.2 Model identity is a claim field

Names alone do not identify a scientific object. The same family label may refer to an upstream release, a local source snapshot, an existing prediction artifact, a retrained adaptation, or a deterministic composition of other artifacts. Table 3 therefore reports the identity that the frozen artifacts support and names fields that remain unknown. This is especially important for Paris: `Paris_CAPER_phase_only` and `TimeXer_local_audited_compact_L168` are development-only local adaptations, not exact transfers of the UrbanEV experts or claims about every upstream implementation.

A.3 Fail-closed state machine

The portable state sequence is:

`REGISTERED` → `RUNNING` → `CLOSED_UNVERIFIED` → `AUDIT_PASS` → `METRIC_UNLOCKED` → `GATE_PASS` or `GATE_FAIL`.

Three side transitions prevent a defective artifact from moving forward. First, an incomplete matrix, target/order mismatch, invalid control, boundary violation, or provenance failure moves `CLOSED_UNVERIFIED` to `BLOCKED` or `REPAIR_REQUIRED`; aggregate metrics remain inaccessible. Second, a permitted repair creates `REPAIRED_UNVERIFIED`, records parent and child hashes and whether predictions changed, and returns to the full audit rather than directly to metric unlock. When selective repair could use information about which controls failed, the formal control family is invalidated and rerun non-selectively. Third, `GATE_FAIL` is terminal for the current protocol version. Continuing with a changed model, threshold, loss, or evidence role requires a new versioned protocol and a valid evidence boundary; a failed version is not rewritten as pending.

The two observed incidents instantiate these transitions. In M7-C, the first audit detected three identity permutations at 69/72 cells while `metrics_unlocked=false`; all 24 shuffled controls were invalidated and rerun before the 72/72 audit passed. In Paris, 32 prediction jobs closed without evaluation metrics, missing-target representation was repaired in 24 artifacts (10,236 entries) with predictions unchanged, and the development metrics were unlocked only after the 32/32 prediction audit passed. These are integrity events, not accuracy improvements.

A.4 Portable lock schema and invariants

The project evolved several lock and receipt formats rather than one universal schema. Table 6 gives the normalized interface required to reproduce the paper’s claim-control semantics. It is a portable schema, not a statement that every historical artifact contains every field. A missing field must be encoded as `unknown` or fail the relevant gate; it must never be silently inferred from a filename.

Table 6: Normalized lock schema. Existing exemplars use heterogeneous field names; the rightmost column states the invariant a portable implementation must enforce.

Group	Required fields	Existing artifact exemplar	Invariant
Protocol	<code>protocol_id</code> , version, status, protocol SHA-256	router, latency, distillation, and Paris teacher protocol files	Data role, endpoint, reference, gate, and stop rule are immutable before aggregate access.
Configuration	<code>run_id</code> , config fingerprint, model identity, seed, fold, horizon	status registry and run manifests	One identifier maps to one configuration; retries retain parentage rather than overwriting identity.
Impl.	code/source hashes; dependency and bundle hashes	Chronos model lock; Paris implementation fingerprints	The implementation that emitted predictions equals the registered implementation or has an explicit non-training amendment receipt.
Environment	Python and library versions; accelerator, runtime, precision; optional environment hash	latency hardware/software manifests; Chronos environment block	Scope is explicit. If no unified environment hash exists, reproducibility is limited rather than implied.
Data boundary	source/data hash, role, permitted metadata, fold dates	baseline protocol; Paris boundary lock and source manifest	Development, formal, and protected roles cannot be silently exchanged.
Target identity	target-value, index/order, time, spatial-unit, and mask hashes; fold-horizon cell prediction SHA-256, shape, finiteness/range, attempt path	expert diagnostics, run status, prediction audits	Candidate and reference use identical target positions and semantics for every compared cell.
Prediction		per-run receipts and prediction audits	The audited bytes are the bytes aggregated; a repair changing predictions requires new scientific evidence.
Metric	metric script/hash, endpoint definition, output hash	metric-lock scripts, aggregate summaries	The metric implementation and weighting rule are fixed; output cannot precede unlock.
Unlock	lock time, conditions, authority, receipt SHA-256, matrix/state hash	Paris metrics-unlock receipt; M7-C formal audit	Every condition is conjunctive and machine-checkable; role-specific unlock does not unlock another role.
Access	analytical-access counter, operation definition, log/receipt location	Paris protected guard and prediction audit	Counter semantics and time origin are stated; storage migration is recorded separately.
Repair	parent hash, child hash, reason, prediction-changed flag, audit return state	shuffled-control amendment and masked-target repair ledger	Repair is traceable and returns to unverified state; non-selective scope is used where observation could bias repair.
Decision	reference, point effect, interval, fold/cell/horizon checks, gate vector, authorized next action	router, Chronos, D1, and Paris teacher summaries/decisions	A failed required component yields terminal failure for that version even when another subcheck passes.
Claim	claim ID, authoritative artifact/hash, evaluation type, allowed/prohibited wording, scope limit	Table 7	Manuscript wording cannot exceed the evidence type or data role.

A.5 Operational definition of analytical access

The Paris guards distinguish three operations.

Storage materialization. Byte-preserving movement, copying, sharding, forensic preservation, and opaque byte hashing of target-bearing files. It also includes creating or verifying a path/size/hash manifest. These operations are recorded as storage events; they do not by themselves authorize or count as target analysis.

Approved metadata handling. Reading the registered non-target fields (date boundary, station identifier, latitude, longitude, postcode, and area), row counts, file sizes, and hashes needed to establish the split and storage receipt, without using state/target values. This is boundary administration, not model evidence.

Analytical access. Any post-lock operation that deserializes, displays, summarizes, plots, filters by, learns from, scores against, or forms a scientific claim from formal/protected target or state values. Manual viewing counts. Feeding values to training, validation, model selection, metric computation, threshold selection, error analysis, or a paper table counts even if no file is modified.

Under this definition, the guard's `analytical_access_count=0` means no protected label was used after the boundary lock for those analytical operations. It does not mean that target bytes were never materialized: one storage-migration event created vault shards and moved recoverable sources. It also does not erase the separately logged display of two example source rows during pre-lock schema discovery. The counter is a procedural project field supported by receipts and audits, not an operating-system reference monitor, append-only security log, encryption guarantee, or defense against a malicious authenticated user.

The in-scope threat is accidental or adaptive analytical use by the project workflow: a script path that reaches a future role, a stale cache, recoverable Git objects in the main project, or a metric opened before the registered closure. The current vault path, manifest, hashes, guards, and negative main-project scan reduce those risks. OS compromise, deliberate bypass by an authorized account, cryptographic confidentiality, and immutable access logging are out of scope. The allowed claim is therefore procedural source separation and zero recorded analytical use, not hardened security.

A.6 Paris fold-synchronous bootstrap, including the partial block

The frozen implementation in `experiments/09_distill_v2/scripts/aggregate_paris_teacher.py` used 5,000 replicates and seed 42. For each development fold it intersected target timestamps across all four horizons, sorted them, required a contiguous hourly sequence, and sliced the sequence at offsets 0, 24, 48, and so on. The final slice was retained even when shorter than 24 hours. Each replicate sampled exactly the fold's number of blocks uniformly with replacement, concatenated the selected timestamp arrays, and reused that same sampled timestamp sequence for every horizon in the fold. Observed-target masks were then applied within each cell, cell RMSEs were macro-averaged, the better CAPER/TimeXer single-model reference was reselected inside the replicate, and teacher gain was computed against that reference. The reported interval is the 2.5th and 97.5th percentiles.

August and October each contain 732 common hourly timestamps, represented as 30 full 24-hour blocks plus one 12-hour block (31 selectable blocks). September and November each contain 708 timestamps, represented as 29 full blocks plus one 12-hour block (30 selectable blocks). The partial block has the same probability of being selected as any other block, but contributes only 12 timestamps whenever selected; consequently replicate timestamp counts can vary with how often the partial block is drawn. No padding, wraparound, discarding, or extra weighting is applied. This exact rule is the scope of the reported Paris interval. No alternative block length or partial-block sensitivity analysis was run for this revision.

A.7 Claim lineage and research-question closure

Table 7 is the manuscript-facing join between a headline result and its machine-readable authority. The hash is computed over the named artifact as stored in the main workspace. A hash identifies bytes; it does not prove that the underlying method is correct. The evaluation-type and wording columns provide the semantic constraint. In particular, the Global Fixed row supports a scoped internal point comparison and a separate measured latency trade-off; it does not acquire an unreported bootstrap interval merely because later failed stages used one.

Table 7: Claim-to-artifact trace and wording guardrails (part 1 of 2). Hashes are the first 12 hexadecimal characters of SHA-256 over the named frozen artifact.

ID	Headline result	Authoritative artifact / SHA-256 ₁₂	Evaluation type	Allowed wording; prohibited wording
C1	Frozen registry: 1,224 configurations; 865 complete artifacts; S1=156.	experiments/00_registry_reclassification/outputs/CONFIG_STATUS_SUMMARY.json 833352c136a4	registry audit	<i>Allowed:</i> In this inventory, completion and exact reproduction differ. <i>Prohibited:</i> A field-wide reproducibility rate or a model-quality ranking.
C2	Source-table status distributions are uneven under the frozen mapping.	experiments/00_registry_reclassification/outputs/CONFIG_STATUS.csv 4b61c2aea510	registry audit	<i>Allowed:</i> Report S1-S6 counts by UrbanEV source table. <i>Prohibited:</i> Call an S4 model invalid or intrinsically weak.
C3	Target-dependent CAPER-TimeXer oracle gain is 16.926%.	experiments/01_expert_diagnostics/outputs/EXPERT_DIAGNOSTIC_TIC_SUMMARY.json 53fa7f22fca6	real-GT oracle diagnostic	<i>Allowed:</i> An infeasible upper bound indicates local complementarity. <i>Prohibited:</i> The oracle is deployable or its gain is recoverable.
C4	State-feature expert-win AUC is 0.6151.	experiments/01_expert_diagnostics/outputs/STATE_FEATURE_DECISION.json 9f84ac483e02	self-supervised proxy	<i>Allowed:</i> Relative expert advantage is moderately predictable. <i>Prohibited:</i> AUC proves lower forecast RMSE or causal demand knowledge.
C5	Strict Global Fixed RMSE is 0.071367, a 3.178% internal gain.	experiments/03_fixed_fusions/outputs/FIXED_FUSION_SUMMARY.json 0a0162204c83	internal rolling real GT	<i>Allowed:</i> Improves on TimeXer under the audited UrbanEV protocol. <i>Prohibited:</i> Blind-test, cross-city, efficient, or deployment-ready.
C6	Direct-loss router gains 0.443% and fails the frozen 1% gate.	experiments/05_router_v1/outputs/ROUTER_V1_SUMMARY.json a9dfc2ebe4a6	internal real GT; gated	<i>Allowed:</i> A positive sub-threshold internal signal; current stage stopped. <i>Prohibited:</i> Router qualification, robustness, or post-hoc gate revision.
C7	Global Fixed batch-1 P50 is 48.86% above TimeXer.	experiments/06_static_latency/LATENCY_SUMMARY_TABLE.csv f94d0a7a9e95	hardware-specific latency	<i>Allowed:</i> Measured on the frozen RTX 4060 FP32 sequential workload. <i>Prohibited:</i> Portable latency, low cost, or universal serving efficiency.

Table 8: Claim-to-artifact trace and wording guardrails (part 2 of 2). Hashes are the first 12 hexadecimal characters of SHA-256 over the named frozen artifact.

ID	Headline result	Authoritative artifact / SHA-256 ₁₂	Evaluation type	Allowed wording; prohibited wording
C8	Chronos-2 gains -3.048% versus Global Fixed and fails qualification.	experiments/06_5_teacher_qualification/outputs/CHRONOS2_QUALIFICATION_SUMMARY.json 01e8084c16ae	zero-shot internal screen	<i>Allowed:</i> This locked revision does not qualify for this protocol. <i>Prohibited:</i> Foundation models are generally inferior for EV data.
C9	Aligned KD gains 0.119% versus identical GT-only; CI crosses zero.	experiments/07_residual_distillation/outputs/D1_SUMMARY.json c38163157c89	internal real GT; gated	<i>Allowed:</i> The primary distillation comparison fails; distill_v1 stops. <i>Prohibited:</i> Distillation improves the student or merits D2 latency.
C10	Aligned KD gains 0.259% versus shuffled teacher.	experiments/07_residual_distillation/outputs/D1_SUMMARY.json c38163157c89	negative control	<i>Allowed:</i> Evidence of a small sample-aligned teacher signal. <i>Prohibited:</i> Engineering value or a passed primary KD claim.
C11	Paris teacher gains 0.847%, wins 4/4 folds and 16/16 cells, but fails 1%.	experiments/09_distill_v2/outputs/teacher_qualification/aggregate/PARIS_TEACHER_QUALIFICATION_SUMMARY.json 5b7c1ddc353c	development real GT	<i>Allowed:</i> Consistent development gain below the frozen qualification effect. <i>Prohibited:</i> Round to 1%, imply formal/protected performance, or call it official TimeXer.
C12	Three identity shuffles were blocked at 69/72 before metric unlock.	experiments/07_residual_distillation/outputs/FORMAL_MATR_IX_AUDIT_V1_0_FAIL_IDENTITY_SHUFFLE.json 584313534390	artifact integrity	<i>Allowed:</i> The lock forced a non-selective rerun of all 24 controls. <i>Prohibited:</i> Use the invalid controls as performance evidence.
C13	Paris prediction audit passed 32/32 before development-metric unlock.	experiments/09_distill_v2/outputs/teacher_qualification/PARIS_TEACHER_PREDICTION_AUDIT.json ee882d8e907b	artifact integrity	<i>Allowed:</i> Identity checks authorized development-metric aggregation. <i>Prohibited:</i> The audit proves model correctness or improved RMSE.
C14	Protected analytical-access counter is 0; vault materialization is true.	experiments/08_paris_new_evidence/PROTECTED_ACCESS_GUARD.json a7780e39f4ef	procedural data boundary	<i>Allowed:</i> No protected label was used analytically after the boundary lock. <i>Prohibited:</i> Targets never existed, were never materialized, or are cryptographically inaccessible.

A.8 Latency across the frozen batch grid

The main text uses batch-one end-to-end P50 as its primary engineering endpoint. Table 9 exposes the complete batch 1/8/32 grid already present in the frozen latency summary. It is generated mechanically by `experiments/10_audit_paper/scripts/generate_revision_appendix.py`; the generator checks for all 18 method–batch rows and formats the existing values without launching the benchmark. The hardware and software context remains the one reported in the original manifests: Windows 10 build 26200, 16 physical/32 logical AMD CPU cores, approximately 16.8 GB RAM, RTX 4060 Laptop GPU with driver 595.71, Python 3.10.8, Torch 2.5.1+cu121, CUDA 12.1, cuDNN 9.1, FP32, and single-threaded Torch/OMP/MKL host settings. GPU clocks were observed per process, not pinned; model loading and checkpoint I/O were excluded. Cold-start, quantized, compiled, parallel-expert, and serverless claims remain outside scope.

Table 9: Frozen end-to-end synchronous latency at batch sizes 1, 8, and 32. P50 and P95 are milliseconds per request; throughput is forecast origins per second at P50. Every value is formatted directly from `LATENCY_SUMMARY_TABLE.csv`; no benchmark was rerun.

Method	Batch origins	E2E P50 (ms)	E2E P95 (ms)	Origins s^{-1} (P50)
DLinear	1	1.166	1.763	857.9
DLinear	8	2.118	2.874	3776.9
DLinear	32	5.382	7.728	5945.7
NLinear	1	0.674	1.056	1483.7
NLinear	8	1.554	2.101	5148.0
NLinear	32	3.896	5.373	8213.7
GBDT-Forecaster	1	6.242	7.195	160.2
GBDT-Forecaster	8	40.188	50.261	199.1
GBDT-Forecaster	32	157.087	186.205	203.7
CAPER	1	7.934	10.798	126.0
CAPER	8	10.740	14.253	744.9
CAPER	32	19.753	24.266	1620.0
TimeXer	1	14.067	15.952	71.1
TimeXer	8	105.780	113.816	75.6
TimeXer	32	482.165	519.827	66.4
Global Fixed CAPER+TimeXer	1	20.940	33.123	47.8
Global Fixed CAPER+TimeXer	8	125.350	136.947	63.8
Global Fixed CAPER+TimeXer	32	479.634	510.574	66.7

Workload: one request produces all four horizons for 275 UrbanEV zones. Measurements used FP32, one RTX 4060 Laptop GPU, sequential expert execution, five processes, 50 warm-ups, and 1,000 timed iterations per process and method–batch cell. Throughput is not monotone across model families because the frozen serving graphs batch differently; these values are workload-specific rather than portable deployment guarantees.

A.9 Protocol and environment artifact index

The following index identifies existing controls; it does not upgrade an incomplete field into a complete reproducibility package. The latency and Chronos stages have explicit environment records. Other stages retain protocol, source/configuration fingerprints, and run receipts but do not expose one uniform cross-stage environment hash, so exact environment equivalence is unknown/not claimed where absent.

Table 10: Protocol and environment artifact index (part 1 of 2).

Artifact	Function	Evidentiary limit
experiments/00_registry_reclassification/scripts/rebuild_status_registry.py experiments/00_registry_reclassification/outputs/CONFIG_STATUS.csv experiments/02_lightweight_baselines/configs/BASELINE_PROTOCOL.json refine-logs/ROUTER_PROTOCOL_V1_0.md and experiments/05_router_v1/configs/ROUTER_V1.json experiments/06_static_latency/LATENCY_PROTOCOL_V1_0.md	Deterministic S1–S6 implementation and priority order Row-level run IDs, fingerprints, grades, reasons, and rerun policy UrbanEV target, folds, horizons, seed, and candidate protocol Router feature/training configuration and frozen Gate 4 Request shape, synchronization, warm-up, repetitions, and primary endpoint	Single programmed classifier; no inter-rater validation One frozen UrbanEV inventory Candidate-set internal evaluation only One seed; proxy and real-GT roles remain separate Warm synchronous serving, not cold start
experiments/06_static_latency/hardware_manifest.json and experiments/06_static_latency/software_manifest.json experiments/06_static_latency/LATENCY_SUMMARY_TABLE.csv	Device, driver, platform, runtime, libraries, precision, host threads Batch 1/8/32 model-only and end-to-end distributions, throughput, memory	One laptop system; clocks observed rather than pinned Hardware/workload-specific
experiments/06_5_teacher_qualification/TEACHER_QUALIFICATION_PROTOCOL_V1_0.md experiments/06_5_teacher_qualification/CHRONOS2_MODEL_LOCK.json experiments/07_residual_distillation/DISTILL_PROTOCOL_V1_0.md experiments/07_residual_distillation/configs/FORMAL_STUDENT_CONFIG_V1.json experiments/07_residual_distillation/SHUFFLE_CONTROL_SPEC.json	Chronos-2 qualification endpoint and gate Resolved model commit, file hashes, snapshot bytes, and environment Teacher, student, loss, controls, milestones, stop rules 338-parameter student input/architecture/training identity 24-hour block-shuffle negative-control construction	Exact project protocol only One revision and zero-shot profile Distill v1 only; later search prohibited after failure One architecture and seed Control test, not a second primary endpoint

Table 11: Protocol and environment artifact index (part 2 of 2).

Artifact	Function	Evidentiary limit
experiments/07_residual_distillation/outputs/FORMAL_MATRIX_AUDIT_V1_0_FAIL_IDENTITY_SHUFFLE.json and experiments/07_residual_distillation/outputs/FORMAL_MATRIX_AUDIT.json experiments/07_residual_distillation/outputs/D1_SUMMARY.json and experiments/07_residual_distillation/outputs/D1_DECISION.md experiments/08_paris_new_evidence/DATA_BOUNDARY_LOCK.json and experiments/08_paris_new_evidence/PROTECTED_ACCESS_GUARD.json experiments/08_paris_new_evidence/PARIS_EVIDENCE_VAULT_REPORT.md experiments/09_distill_v2/PARIS_TEACHER_QUALIFICATION_PROTOCOL_V1_0.md experiments/09_distill_v2/PARIS_TEACHER_MODEL_CONFIG.json experiments/09_distill_v2/TIMEXER_LOCAL_ADAPTATION_PROVENANCE.json and experiments/09_distill_v2/PARIS_CAPER_ADAPTATION_V1_0.md experiments/09_distill_v2/PARIS_TEACHER_DEPLOY_APPROVAL_V1_1.md experiments/09_distill_v2/outputs/teacher_qualification/PARIS_TEACHER_PREDICTION_AUDIT.json and experiments/09_distill_v2/outputs/teacher_qualification/PARIS_TEACHER_METRICS_UNLOCK_RECEIPT.json experiments/09_distill_v2/outputs/teacher_qualification/aggregate/PARIS_TEACHER_QUALIFICATION_SUMMARY.json experiments/09_distill_v2/PARIS_TEACHER_RESULT_REVIEW.md experiments/10_audit_paper/scripts/generate_revision_appendix.py	Pre-unlock failure and final 72/72 audit Primary/negative-control effects and terminal gate decision Role dates, permitted metadata, storage state, access counter, unlock rule Vault identities, hashes, negative main-project scan, threat limits Development folds, target, missingness, time rule, teacher fit, gate Local CAPER/TimeXer profiles, training, queue, bootstrap Source hash and explicit local adaptation differences Automated differential audit closure after the earlier block 32/32 closure, identities, role-specific metric unlock, access counts Point effect, fold/cell results, synchronized bootstrap, gate vector Separately instantiated result-to-claim check and provenance warnings Mechanical hash/latency table generation	Supports control integrity, not accuracy No D2 latency, multi-seed, or protected evidence Procedural guard; not hardened authorization Metadata/hash audit only; no content-level protected audit Paris development only Local adaptations, not exact upstream implementations Upstream TimeXer commit undisclosed; Paris CAPER newly trained Automated artifact audit, not blinded human replication Development-metric authorization only Current hypothesis failed; no student/formal/protected result Warnings limit stronger process claims; they do not reverse the gate No model, metric, resampling, or vault operation

A.10 Author and reviewer application checklist

Table 12 turns the case-study controls into a review interface. The final row is important: missing validation must be stated as missing. This paper therefore treats the 1% rule as a frozen local convention without an economic utility model, the S1–S6 labels as deterministic without inter-rater evidence, and all model/latency conclusions as bounded to their recorded seeds, panels, adaptations, and serving graph.

Table 12: Portable author and reviewer checklist for auditable forecasting claims.

ID	Check	Author must expose	Reviewer test
1	Target and role semantics	numerator/denominator, spatial unit, mask, sampling clock, and development/formal/protected role	Could two datasets or target constructions be mistaken for the same RMSE scale?
2	Configuration identity	run ID, protocol version, fingerprint, source mapping, seed, fold/horizon, and execution state	Could a completed run still be underidentified or map to several disclosed cells?
3	Information timing	forecast origin, lookback, covariates, fill/scaler fit interval, and prequential weight/tuning source	Does any current/future target influence a parameter evaluated in the same fold?
4	Artifact closure	expected matrix cardinality, terminal attempts, prediction/target/order/time/spatial/mask identities	Are aggregates withheld until every required cell is closed and verified?
5	Metric lock	metric implementation identity, lock state, unlock conditions, authority, receipt, and access counter	Can any result be read or selected before the audit passes?
6	Comparator and endpoint	primary reference, macro/micro weighting, endpoint, sign convention, and secondary diagnostics	Was the reference or endpoint changed after seeing results?
7	Uncertainty and gate	resampling unit, partial-block rule, replicates/seed, minimum effect, consistency checks, stop rule	Is a positive interval against zero being substituted for a failed effect-size requirement?
8	Controls and repair	negative-control construction, invalidation scope, before/after hashes, and whether predictions changed	Were only visibly bad cells repaired, or was the formal control family rerun non-selectively?
9	Model identity	upstream/local source, checkpoint/revision, local adaptations, input contract, and every unknown field	Does a local adaptation masquerade as an official or exact upstream implementation?
10	Engineering measurement	device, software, precision, batch/request shape, synchronization, warm-up, repetitions, P50/P95, throughput	Is a hardware-specific measurement being promoted to portable efficiency?
11	Future/protected evidence	storage versus analytical access, allowed metadata, threat-model limit, logs, and unlock rule	Does the boundary prevent analysis, merely record storage, or claim stronger security than implemented?
12	Claim lineage	claim ID, authoritative artifact, SHA-256, evaluation type, allowed/prohibited wording, scope limit	Can every headline number be traced, and are diagnostics/subchecks kept at their original evidence type?
13	Negative result	terminal decision, disallowed same-interval actions, and requirements for a new version/evidence boundary	Was a failed hypothesis silently continued through tuning, threshold changes, or a new structure?
14	Missing validation	limitations must name absent multi-seed, multi-hardware, external-team, utility, or rater evidence	Is any absent analysis implied by rhetoric such as robust, validated, secure, or general?

References

- [1] Han Li, Haohao Qu, Xiaojun Tan, Linlin You, Rui Zhu, and Wenqi Fan. UrbanEV: An open benchmark dataset for urban electric vehicle charging demand prediction. *Scientific Data*, 12(1):523, 2025. doi: 10.1038/s41597-025-04874-4. URL <https://doi.org/10.1038/s41597-025-04874-4>.
- [2] Zihan Guo, Linlin You, Rui Zhu, Yan Zhang, and Chau Yuen. A city-scale and harmonized dataset for global electric vehicle charging demand analysis. *Scientific Data*, 12(1):1254, 2025. doi: 10.1038/s41597-025-05584-7. URL <https://doi.org/10.1038/s41597-025-05584-7>.
- [3] Yvenn Amara-Ouali, Yannig Goude, Nathan Doumèche, Pascal Veyret, Alexis Thomas, Daniel Hebenstreit, Thomas Wedenig, Arthur Satouf, Aymeric Jan, Yannick Deleuze, Paul Berhaut, and Sebastien Treguer. Forecasting electric vehicle charging station occupancy: Smarter mobility data challenge. *Journal of Data-centric Machine Learning Research*, 1(16):1–27, 2024. URL <https://openreview.net/forum?id=OwOwrH5fXQ>.
- [4] Yvenn Amara-Ouali, Yannig Goude, Nathan Doumèche, Pascal Veyret, Alexis Thomas, Daniel Hebenstreit, Thomas Wedenig, Arthur Satouf, Aymeric Jan, Yannick Deleuze, Paul Berhaut, and Sébastien Treguer. Electric vehicle occupancy of charging points in the city of paris, 2023. URL <https://doi.org/10.5281/zenodo.8280566>. Dataset.
- [5] Zachary J. Lee, Tongxin Li, and Steven H. Low. ACN-Data: Analysis and applications of an open EV charging dataset. In *Proceedings of the Tenth ACM International Conference on Future Energy Systems*, pages 139–149, 2019. doi: 10.1145/3307772.3328313. URL <https://doi.org/10.1145/3307772.3328313>.
- [6] Haohao Qu, Haoxuan Kuang, Qiuxuan Wang, Jun Li, and Linlin You. A physics-informed and attention-based graph learning approach for regional electric vehicle charging demand prediction. *IEEE Transactions on Intelligent Transportation Systems*, 25(10):14284–14297, 2024. doi: 10.1109/TITS.2024.3401850. URL <https://doi.org/10.1109/TITS.2024.3401850>.
- [7] Haohao Qu, Han Li, Linlin You, Rui Zhu, Jinyue Yan, Paolo Santi, Carlo Ratti, and Chau Yuen. ChatEV: Predicting electric vehicle charging demand as natural language processing. *Transportation Research Part D: Transport and Environment*, 136:104470, 2024. doi: 10.1016/j.trd.2024.104470. URL <https://doi.org/10.1016/j.trd.2024.104470>.
- [8] Yuxuan Wang, Haixu Wu, Jiayang Dong, Guo Qin, Haoran Zhang, Yong Liu, Yunzhong Qiu, Jianmin Wang, and Mingsheng Long. TimeXer: Empowering transformers for time series forecasting with exogenous variables. In *Advances in Neural Information Processing Systems*, volume 37, pages 469–498, 2024. doi: 10.52202/079017-0015. URL https://proceedings.neurips.cc/paper_files/paper/2024/hash/0113ef4642264adc2e6924a3cbbdf532-Abstract-Conference.html.
- [9] Ailing Zeng, Muxi Chen, Lei Zhang, and Qiang Xu. Are transformers effective for time series forecasting? *Proceedings of the AAAI Conference on Artificial Intelligence*, 37(9):11121–11128, 2023. doi: 10.1609/aaai.v37i9.26317. URL <https://doi.org/10.1609/aaai.v37i9.26317>.
- [10] Abdul Fatir Ansari, Lorenzo Stella, Ali Caner Türkmen, Xiyuan Zhang, Pedro Mercado, Huibin Shen, Oleksandr Shchur, Syama Sundar Rangapuram, Sebastian Pineda Arango, Shubham Kapoor, Jasper Zschiegner, Danielle C. Maddix, Hao Wang, Michael W. Mahoney, Kari Torkkola, Andrew Gordon Wilson, Michael Bohlke-Schneider, and Bernie Wang. Chronos: Learning the language of time series. *Transactions on Machine Learning Research*, 2024. ISSN 2835-8856. URL <https://openreview.net/forum?id=gerNCVqqtR>.
- [11] Abdul Fatir Ansari, Oleksandr Shchur, Jaris Küken, Andreas Auer, Boran Han, Pedro Mercado, Syama Sundar Rangapuram, Huibin Shen, Lorenzo Stella, Xiyuan Zhang, Mononito Goswami, Shubham Kapoor, Danielle C. Maddix, Pablo Gueron, Tony Hu, Junming Yin, Nick Erickson, Prateek Mutalik Desai, Hao Wang, Huzefa Rangwala, George Karypis, Yuyang Wang, and Michael Bohlke-Schneider. Chronos-2: From univariate to universal forecasting. *arXiv preprint arXiv:2510.15821*, 2025. URL <https://arxiv.org/abs/2510.15821>.
- [12] A. P. Dawid. Present position and potential developments: Some personal views: Statistical theory: The prequential approach. *Journal of the Royal Statistical Society. Series A (General)*, 147(2):278–290, 1984. doi: 10.2307/2981683. URL <https://doi.org/10.2307/2981683>.
- [13] Leonard J. Tashman. Out-of-sample tests of forecasting accuracy: An analysis and review. *International Journal of Forecasting*, 16(4):437–450, 2000. doi: 10.1016/S0169-2070(00)00065-0. URL [https://doi.org/10.1016/S0169-2070\(00\)00065-0](https://doi.org/10.1016/S0169-2070(00)00065-0).
- [14] Christoph Bergmeir and José M. Benítez. On the use of cross-validation for time series predictor evaluation. *Information Sciences*, 191:192–213, 2012. doi: 10.1016/j.ins.2011.12.028. URL <https://doi.org/10.1016/j.ins.2011.12.028>.

- [15] Vitor Cerqueira, Luis Torgo, and Igor Mozetič. Evaluating time series forecasting models: An empirical study on performance estimation methods. *Machine Learning*, 109(11):1997–2028, 2020. doi: 10.1007/s10994-020-05910-7. URL <https://doi.org/10.1007/s10994-020-05910-7>.
- [16] Hansika Hewamalage, Klaus Ackermann, and Christoph Bergmeir. Forecast evaluation for data scientists: Common pitfalls and best practices. *Data Mining and Knowledge Discovery*, 37(2):788–832, 2023. doi: 10.1007/s10618-022-00894-5. URL <https://doi.org/10.1007/s10618-022-00894-5>.
- [17] Rob J. Hyndman and Anne B. Koehler. Another look at measures of forecast accuracy. *International Journal of Forecasting*, 22(4):679–688, 2006. doi: 10.1016/j.ijforecast.2006.03.001. URL <https://doi.org/10.1016/j.ijforecast.2006.03.001>.
- [18] Spyros Makridakis, Evangelos Spiliotis, and Vassilios Assimakopoulos. M5 accuracy competition: Results, findings, and conclusions. *International Journal of Forecasting*, 38(4):1346–1364, 2022. doi: 10.1016/j.ijforecast.2021.11.013. URL <https://doi.org/10.1016/j.ijforecast.2021.11.013>.
- [19] Gavin C. Cawley and Nicola L. C. Talbot. On over-fitting in model selection and subsequent selection bias in performance evaluation. *Journal of Machine Learning Research*, 11:2079–2107, 2010. URL <https://www.jmlr.org/papers/v11/cawley10a.html>.
- [20] Cynthia Dwork, Vitaly Feldman, Moritz Hardt, Toniann Pitassi, Omer Reingold, and Aaron Roth. The reusable holdout: Preserving validity in adaptive data analysis. *Science*, 349(6248):636–638, 2015. doi: 10.1126/science.aaa9375. URL <https://doi.org/10.1126/science.aaa9375>.
- [21] Sayash Kapoor and Arvind Narayanan. Leakage and the reproducibility crisis in machine-learning-based science. *Patterns*, 4(9):100804, 2023. doi: 10.1016/j.patter.2023.100804. URL <https://doi.org/10.1016/j.patter.2023.100804>.
- [22] Joelle Pineau, Philippe Vincent-Lamarre, Koustuv Sinha, Vincent Larivière, Alina Beygelzimer, Florence d’Alché Buc, Emily Fox, and Hugo Larochelle. Improving reproducibility in machine learning research: A report from the NeurIPS 2019 reproducibility program. *Journal of Machine Learning Research*, 22(164):1–20, 2021. URL <https://jmlr.org/papers/v22/20-303.html>.
- [23] Xavier Bouthillier, Pierre Delaunay, Mirko Bronzi, Assya Trofimov, Brennan Nichyporuk, Justin Szeto, Nazanin Mohammadi Sepahvand, Edward Raff, Kanika Madan, Vikram Voleti, Samira Ebrahimi Kahou, Vincent Michalski, Tal Arbel, Chris Pal, Gaël Varoquaux, and Pascal Vincent. Accounting for variance in machine learning benchmarks. In *Proceedings of Machine Learning and Systems*, volume 3, pages 747–769, 2021. URL https://proceedings.mlsys.org/paper_files/paper/2021/hash/0184b0cd3cfb185989f858a1d9f5c1eb-Abstract.html.
- [24] Yvenn Amara-Ouali, Yannig Goude, Pascal Massart, Jean-Michel Poggi, and Hui Yan. A review of electric vehicle load open data and models. *Energies*, 14(8):2233, 2021. doi: 10.3390/en14082233. URL <https://doi.org/10.3390/en14082233>.
- [25] Mamunur Rashid, Tarek Elfouly, and Nan Chen. A comprehensive survey of electric vehicle charging demand forecasting techniques. *IEEE Open Journal of Vehicular Technology*, 5:1348–1373, 2024. doi: 10.1109/OJVT.2024.3457499. URL <https://doi.org/10.1109/OJVT.2024.3457499>.
- [26] Luboš Buzna, Pasquale De Falco, Gabriella Ferruzzi, Shahab Khormali, Daniela Proto, Nazir Refa, Milan Straka, and Gijs van der Poel. An ensemble methodology for hierarchical probabilistic electric vehicle load forecasting at regular charging stations. *Applied Energy*, 283:116337, 2021. doi: 10.1016/j.apenergy.2020.116337. URL <https://doi.org/10.1016/j.apenergy.2020.116337>.